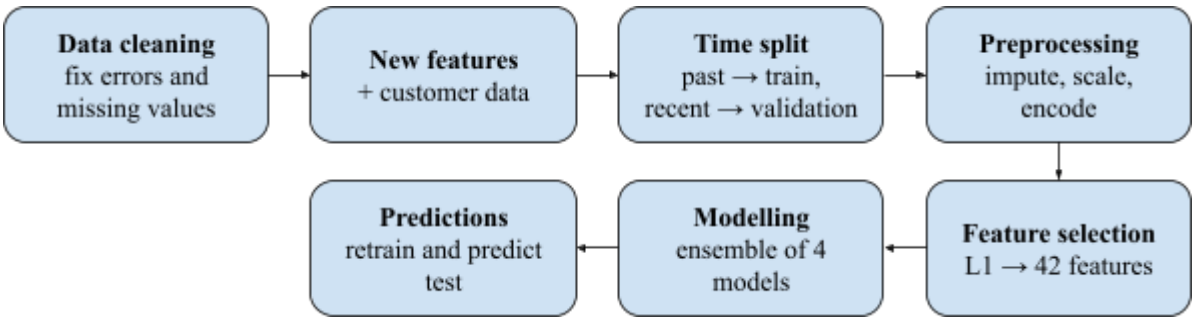
Introduction

In this project we explore a dataset from Everything & Then Some, a wholesale company. Our goal is to create a binary classification model that predicts whether a customer will make another purchase within 90 days of the current purchase (repeat_purchase_90d). We analysed 6,534 historical purchases and retained 6,469 eligible observations for model training. We then generated predictions for 986 later purchases, incorporating additional customer information from clientes.csv. This project includes data exploration, pre-processing, feature selection, model training and evaluation, and the generation of the final predictions.

Pipeline Overview



Data Cleaning

Our first step was observing and cleaning the training and the test datasets, always applying the same rules to both so that no information from the test set was used. We found 65 training rows that contained only the identifiers and the target, without any purchase information, and we removed them. Many missing values were not errors but a natural consequence of the customer's history: a customer with fewer than two previous purchases cannot have an average time between purchases or a standard deviation of purchase values. These values were set to zero, and we created indicator variables so that the model could still recognise customers with little or no purchase history. We also identified several invalid values, including negative purchase intervals, negative annual spending, and an unusually large placeholder value of 1,000,000,000. These were treated as missing values, and the average purchase value was recalculated using total spending and the number of previous purchases.

Data Exploration and Preprocessing

The exploration of the training data revealed several important patterns. The target classes were fairly balanced, with 56% of purchases followed by a repeat purchase. The strongest pattern was related to purchase behaviour: customers who buy more often and more recently are much more likely to return, while the value of the current purchase has very little influence. We also found that monetary values and purchase counts were highly skewed, and that some variables contained overlapping information, such as the different purchase totals. Since the test purchases took place after the training period, the test data represents a later period. We also observed that test customers tended to be more active, with higher levels of purchasing activity.

From the customer table (clientes.csv) we added the commercial region and the year in which each customer registered, since the remaining columns either described individual customers or were empty. Identifiers and the random_noise variable were excluded from the model. We also created new features, such as the purchase_month and purchase_weekday, recent_spend_share, and purchase_gap_ratio (a ratio that compares the time since the last purchase with the customer's usual time between purchases).
Because the test set lies in the future, we split the training data by date rather than randomly: the older purchases were used for training and the most recent ones for validation, with a 90-day gap between them to prevent the target periods from overlapping. Missing values were filled with the median of the training data, numerical variables were standardised with the StandardScaler, since our feature selection and Logistic Regression compare coefficients whose size depends on the scale of each variable, and categorical variables were transformed through One-Hot Encoding, with rare categories grouped together. All these transformations were learned from the training data only.

Feature Selection
After preprocessing, we performed feature selection to identify the most relevant variables for predicting repeat purchases. We compared three approaches: SelectKBest with the ANOVA F-test, Recursive Feature Elimination (RFE), and L1-regularized Logistic Regression, as well as a voting approach that combined their results. Each feature set was evaluated with the same Logistic Regression on the validation data. Although RFE achieved a slightly higher initial F1-score, L1 regularisation achieved the highest ROC-AUC of 0.81, and it also removed some redundant variables found during the exploration. We then removed a few salesperson and transport-method categories that appeared to behave like identifiers, leaving 42 features with almost no change in ROC-AUC. When we later trained with alternative models, the 42-feature subset generally performed better for most models than both the full set of 137 features and the subset selected through RFE, supporting our decision to use the reduced L1 feature-selection approach.

Model Training and Evaluation
We compared different classification models, including: Logistic Regression, Random Forest, Extra Trees and Gradient Boosting. The models were evaluated using a chronological train-validation split, which allowed us to assess how well the models were predicting future purchases while reducing the risk of temporal leakage. We tuned hyperparameters using ROC-AUC and selected a soft-voting ensemble combining the four models. We optimized the classification threshold using F1-score, retrained the ensemble on 6,469 observations and generated 986 test predictions, preserving the original IDs. After several iterations, our latest Kaggle submission achieved an F1-score of 0.79899.

Conclusion
In this project, we explored different approaches to predict repeat purchases within 90 days. We developed a soft-voting ensemble by combining four classification models. We also saw how changes in customer behaviour can make predictions more challenging and why careful model evaluation is important.